

MTH 203 : Lecture 02

Ex: Show that the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

is divergent.

Solⁿ: Let us write the terms of the series as:

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4} \right)}_{= \frac{7}{12} > \frac{6}{12} = \frac{1}{2}} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \right)}_{> \frac{4}{8} = \frac{1}{2}} + \underbrace{\left(\frac{1}{9} + \frac{1}{10} + \dots + \frac{1}{16} \right)}_{> \frac{8}{16} = \frac{1}{2}} + \dots$$

So, the sum of 2^n terms ending with $\frac{1}{2^{n+1}}$ is greater than $\frac{2^n}{2^{n+1}} = \frac{1}{2}$

Thus if (s_n) is the sequence of partial sum of the series then

$$s_{2^1} = s_2 = 1 + \frac{1}{2} = \frac{3}{2} > \frac{1}{2}$$

$$s_{2^2} = s_4 > \frac{2}{2} = 1$$

$$s_{2^3} = s_8 > \frac{3}{2}$$

$$s_{2^4} = s_{16} > \frac{4}{2}$$

In general $s_{2^k} > \frac{k}{2}$

So, the (s_n) is an increasing sequence which diverges.

Hence $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

The Integral Test :

Theorem: Let $\{a_n\}$ be a sequence of positive terms. Suppose $a_n = f(n)$ where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x) dx$ both converge or both diverge.

Ex: Consider the p series
$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} + \dots$$
where p is a real constant.

Solⁿ: Consider $f(x) = \frac{1}{x^p}$
If $p > 1$, f is a positive decreasing function of x .

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^p} dx &= \int_1^{\infty} x^{-p} dx = \lim_{b \rightarrow \infty} \left[\frac{x^{-p+1}}{-p+1} \right]_1^b \\ &= \frac{1}{1-p} [0 - 1] = \frac{1}{p-1} < \infty \\ &\quad (\text{if } p > 1) \end{aligned}$$

So, $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$.

For $p < 1$, $\int_1^{\infty} \frac{1}{x^p} dx = \frac{1}{(1-p)} \lim_{b \rightarrow \infty} (b^{1-p} - 1)$
 $= \infty$ (since $p < 1$)

So, if $p < 1$, the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges.

Ex: Consider $\sum_{n=1}^{\infty} \frac{8 \tan^{-1} n}{1+n^2}$

consider $f(x) = \frac{8 \tan^{-1} x}{1+x^2}$

$$\begin{aligned} \text{Now } \int_1^{\infty} \frac{8 \tan^{-1} x}{1+x^2} dx &= \int_{\pi/4}^{\pi/2} 8u du = \left[8 \frac{u^2}{2} \right]_{\pi/4}^{\pi/2} \\ &= 4 \left(\frac{\pi^2}{4} - \frac{\pi^2}{16} \right) \\ &= 4 \times \frac{3\pi^2}{16} = \frac{3\pi^2}{4} < \infty \end{aligned}$$

Let $u = \tan^{-1} x$
 $du = \frac{1}{1+x^2} dx$

Hence the series $\sum_{n=1}^{\infty} \frac{8 \tan^{-1} n}{1+n^2}$ Converges.

Ex:
$$\sum_{n=1}^{\infty} \frac{2 + (-1)^n}{(1.25)^n}$$

Now
$$\frac{2 + (-1)^n}{(1.25)^n} = \frac{2 + (-1)^n}{\left(\frac{5}{4}\right)^n} = \left(\frac{4}{5}\right)^n [2 + (-1)^n] \leq \left(\frac{4}{5}\right)^n \cdot 3$$

Now $\left(\frac{4}{5}\right)^n$ is the n th term of a convergent geometric series

and so $3\left(\frac{4}{5}\right)^n$ is also the n th term of a convergent series.

So the given series converges by Comparison test.

Ex:
$$\sum_{n=1}^{\infty} \frac{\ln(n)}{n}$$

Note that $\frac{\ln(n)}{n} > \frac{1}{n}$ for $n \geq 3$

Since $\sum \frac{1}{n}$ is a divergent series, by comparison test, the given series diverges.

Ex: $\sum_{n=1}^{\infty} \frac{\tanh(n)}{n^2}$

Let us compare with $\sum_{n=1}^{\infty} \frac{1}{n^2}$

$$\lim_{n \rightarrow \infty} \left(\frac{\frac{\tanh(n)}{n^2}}{\frac{1}{n^2}} \right) = \lim_{n \rightarrow \infty} (\tanh(n)) = \lim_{n \rightarrow \infty} \frac{e^n - e^{-n}}{e^n + e^{-n}}$$

$$= \lim_{n \rightarrow \infty} \frac{1 - e^{-2n}}{1 + e^{-2n}} = 1 \neq 0$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, by Limit Comparison test, the given series converges.

Ex: $\sum_{n=1}^{\infty} \frac{1}{n \sqrt[n]{n}}$

Let us compare with the series $\sum_{n=1}^{\infty} \frac{1}{n}$

$$\lim_{n \rightarrow \infty} \left(\frac{\frac{1}{n \sqrt[n]{n}}}{\frac{1}{n}} \right) = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n}} = 1 \neq 0$$

Since the series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, by Limit Comparison test, the given series also diverges.

Ex. $\sum_{n=1}^{\infty} \frac{1}{(1+2+3+\dots+n)}$

The n th term $= \frac{1}{1+2+\dots+n} = \frac{1}{\frac{n(n+1)}{2}}$

$$= \frac{2}{n(n+1)}$$

Let us compare it with the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$

Now $\lim_{n \rightarrow \infty} \frac{\frac{2}{n(n+1)}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{2n^2}{n(n+1)}$

$$= \lim_{n \rightarrow \infty} \frac{2n}{(n+1)} = \lim_{n \rightarrow \infty} \frac{2}{\left(1 + \frac{1}{n}\right)} = 2 \neq 0$$

Since the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges,

by limit Comparison test, the given series converges.

Theorem: (The Ratio Test):

Let $\sum a_n$ be a series with positive terms
and suppose that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = l$

Then

- (a) The series converges if $l < 1$
- (b) The series diverges if $l > 1$ (or $l = \infty$)
- (c) The test is inconclusive if $l = 1$

Ex: $\sum_{n=1}^{\infty} \frac{n 2^n (n+1)!}{3^n n!}$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1) \overset{2}{\cancel{2^{n+1}}} \overset{(n+2)}{\cancel{(n+2)!}}}{\cancel{3^{n+1}} \overset{(n+1)}{\cancel{(n+1)!}}} \times \frac{\cancel{3^n} \cancel{n!}}{n \cancel{2^n} \cancel{(n+1)!}}$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)}{n} \cdot \frac{2}{3} \cdot \frac{(n+2)}{(n+1)}$$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) \cdot \frac{2}{3} \cdot \frac{(1 + 2/n)}{(1 + 1/n)} = \frac{2}{3} < 1$$

So, by Ratio test the series converges.

Ex: $\sum_{n=1}^{\infty} \frac{3^n}{n^3 2^n}$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{3^{n+1}}{(n+1)^3 2^{n+1}} \cdot \frac{n^3 2^n}{3^n} = \lim_{n \rightarrow \infty} \frac{3}{2} \cdot \frac{n^3}{(n+1)^3}$$

$$= \lim_{n \rightarrow \infty} \frac{3}{2} \cdot \frac{1}{\left(1 + \frac{1}{n}\right)^3} = \frac{3}{2} > 1$$

So, by Ratio test, the series diverges.

Theorem (The Root test):

Let $\sum a_n$ be a series with $a_n \geq 0$ for $n \geq N$ and suppose that $\lim_{n \rightarrow \infty} (a_n)^{1/n} = l$

Then

- (a) The series converges if $l < 1$
- (b) The series diverges if $l > 1$ or $l = \infty$
- (c) The test is inconclusive if $l = 1$.

Ex: $\sum_{n=2}^{\infty} \frac{n}{(\log n)^{n/2}}$

$$\begin{aligned} \lim_{n \rightarrow \infty} (a_n)^{1/n} &= \lim_{n \rightarrow \infty} \left\{ \frac{n}{(\log n)^{n/2}} \right\}^{1/n} = \lim_{n \rightarrow \infty} \frac{(n^{1/n})}{\sqrt{\log n}} \\ &= \frac{\lim_{n \rightarrow \infty} (n^{1/n})}{\lim_{n \rightarrow \infty} \sqrt{\log n}} = 0 < 1 \\ &\quad \left(\text{since } \lim_{n \rightarrow \infty} (n^{1/n}) = 1 \right) \end{aligned}$$

So, by the Root test, the series converges.

Ex: $\sum_{n=1}^{\infty} \frac{2^n}{n^2}$

$$\begin{aligned} \lim_{n \rightarrow \infty} (a_n)^{1/n} &= \lim_{n \rightarrow \infty} \left(\frac{2^n}{n^2} \right)^{1/n} = \lim_{n \rightarrow \infty} \frac{2}{n^{2/n}} \\ &= \lim_{n \rightarrow \infty} \frac{2}{(n^{1/n})^2} = \frac{2}{1} = 2 > 1 \end{aligned}$$

So, by the Root test, the series diverges.

Alternating Series:

A series in which the terms are alternately positive and negative is called an alternating series.

e.g. ① $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{(-1)^{n+1}}{n} + \dots$

(Alternating harmonic series)

② $1 - 2 + 3 - 4 + \dots + (-1)^{n+1} n + \dots$

Leibnitz's Theorem (The Alternating Series Test)

The series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \dots$

converges if all three of the following conditions are satisfied.

(1) The u_n 's are all positive.

(2) $u_n > u_{n+1}$ for all $n \geq N$, for some integer N

(3) $u_n \rightarrow 0$

Ex: The alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{(-1)^{n+1}}{n} + \dots$$

converges by Leibnitz's Theorem.

Absolute and Conditional Convergence:

• Absolutely Convergent: A series $\sum a_n$ converges absolutely (is absolutely convergent) if the corresponding series of absolute values, $\sum |a_n|$ converges.

Ex: The series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n}$$

converges absolutely because $\sum_{n=0}^{\infty} \frac{1}{2^n}$ converges (a geometric series with common ratio $\frac{1}{2}$)

Ex: The alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

does not converge absolutely

because $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

Conditionally Convergent:

A series that converges but does not converge absolutely is called Conditionally Convergent.

Ex: The alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \text{ is Conditionally Convergent.}$$

Theorem (The Absolute Convergence Test):

If $\sum_{n=1}^{\infty} |a_n|$ Converges then $\sum_{n=1}^{\infty} a_n$ Converges.

Ex:
$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^3} = 1 - \frac{1}{8} + \frac{1}{27} - \frac{1}{64} + \dots$$

Converges because the corresponding series of absolute values

$$\sum_{n=1}^{\infty} \frac{1}{n^3} = 1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \dots$$

is a convergent series.

Rearranging Series:

Theorem (The Rearrangement Theorem for Absolutely Convergent Series):

If $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $b_1, b_2, \dots, b_n, \dots$ is any arrangement of the sequence (a_n) then $\sum_{n=1}^{\infty} b_n$ converges absolutely and $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n$

Ex: We have seen that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$ converges absolutely.

Using the above theorem

$$1 - \frac{1}{8} - \frac{1}{64} + \frac{1}{27} + \frac{1}{125} + \frac{1}{243} - \dots$$

converges absolutely and both series converge to the same value.

Note: The alternating harmonic series

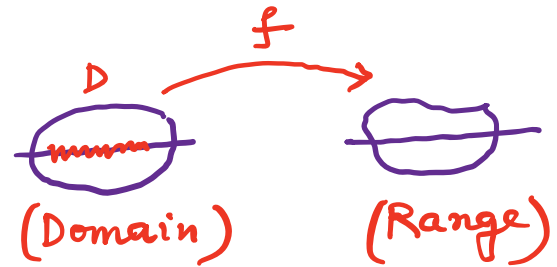
$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \dots$$

can be rearranged to diverge or reach any preassigned term.

[Function of Several Variables]

Recall:

Function of one variable:



Let $D \subseteq \mathbb{R}$

Let $f: D \rightarrow \mathbb{R}$ be a function from D to $\mathbb{R}$

This means f is a rule which assigns to each element $x \in D$, a unique element $y \in \mathbb{R}$.
We write $y = f(x)$.

Here: $f \rightarrow$ The rule, $f(x) \rightarrow$ Value of f at x
(or, image of x under f)

$x \rightarrow$ independent variable

$y \rightarrow$ dependent variable

Domain D : The set of values (points) for which f is defined.

Range: The set of values of f

f is called Real valued function if

$$\text{Range}(f) \subseteq \mathbb{R}$$

Now we want to generalize these concepts to several variables.

Let n be any positive integer;

Then $\mathbb{R}^n = \{ (x_1, x_2, \dots, x_n) : x_i \in \mathbb{R} \text{ for } i=1, 2, \dots, n \}$
(Set of all ordered n -tuples of real numbers)

Now we need a notion of distance on $\mathbb{R}^n$

- First observe that $\mathbb{R}^n$ is a vector space (or linear space) over the field $\mathbb{R}$

So, there are two operations on $\mathbb{R}^n$:

: vector addition

: scalar multiplication

- Furthermore, $\mathbb{R}^n$ is an inner product space.
(vector space with an inner product)

On $\mathbb{R}^n$, the inner product is defined as follows:

If $x = (x_1, x_2, \dots, x_n)$ and $y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$,

then

$$\langle x, y \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n = \sum_{i=1}^n x_i y_i$$

Properties: $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$

For $x, y, z, w \in \mathbb{R}^n$,

- $\langle x, x \rangle \geq 0 \quad \forall x \in \mathbb{R}^n$ and $\langle x, x \rangle = 0 \iff x = 0$

- $\langle z+w, y \rangle = \langle z, y \rangle + \langle w, y \rangle$
- $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle \quad \forall \alpha \in \mathbb{R}$
- $\langle x, y \rangle = \langle y, x \rangle$

Cauchy - Schwarz Inequality:

$$|\langle x, y \rangle| \leq \sqrt{\langle x, x \rangle} \sqrt{\langle y, y \rangle} \quad \forall x, y \in \mathbb{R}^n$$

or equivalently

$$\left| \sum_{i=1}^n x_i y_i \right| \leq \left(\sum_{i=1}^n x_i^2 \right)^{1/2} \left(\sum_{i=1}^n y_i^2 \right)^{1/2}$$

Equality holds $\Leftrightarrow$ x and y are linearly dependent

• (Euclidean) Norm induced by the inner product

If $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$,
define $\|x\| = \sqrt{\langle x, x \rangle}$

So, $\|\cdot\| : \mathbb{R}^n \longrightarrow [0, \infty)$

• Properties :

• $\|x\| \geq 0 \quad \forall x \in \mathbb{R}^n$
and $\|x\| = 0 \Leftrightarrow x = 0$

• $\|\alpha x\| = |\alpha| \|x\| \quad \forall \alpha \in \mathbb{R}$

• $\|x+y\| \leq \|x\| + \|y\|$ (Triangle inequality)

Question : Prove triangle inequality.
When does the equality hold?

$\mathbb{R}^n$, equipped with the inner product
(hence the norm)
defined above is called n -dimensional
Euclidean Space.

Now for any $x, y \in \mathbb{R}^n$, we can define
distance between
 x and $y \quad = \|x-y\| = \left\{ \sum_{i=1}^n (x_i - y_i)^2 \right\}^{1/2}$

Function of n -variables :

Let $D \subseteq \mathbb{R}^n$

A real valued function $f: D \rightarrow \mathbb{R}$ is a rule
which associates to each point $x = (x_1, x_2, \dots, x_n) \in D$,
a unique point $z \in \mathbb{R}$ and is denoted by

$$z = f(x_1, x_2, \dots, x_n)$$

The rule

Independent variables

The value of f

at $x = (x_1, x_2, \dots, x_n)$

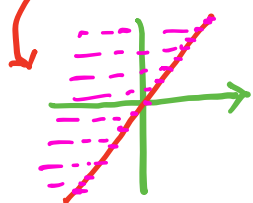
= Image of x under f = Dependent variable

Domain D : Set of points where f is defined.

Range: Set of values of f .

Ex ①: $f(x, y) = \sqrt{y-x}$

Domain = $\{(x, y) \in \mathbb{R}^2 : y \geq x\}$



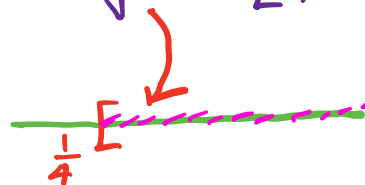
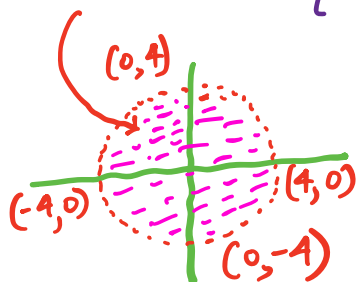
Range = $[0, \infty)$



Ex ②: $f(x, y) = \frac{1}{\sqrt{16-x^2-y^2}}$

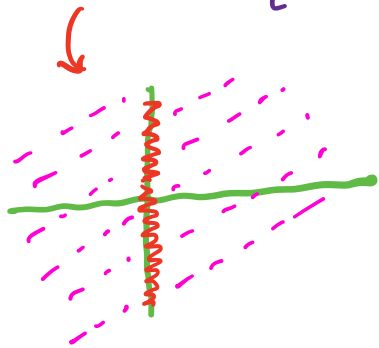
f is defined when $16-x^2-y^2 > 0$
 $\Rightarrow x^2+y^2 < 16$

Domain = $\{(x, y) \in \mathbb{R}^2 : x^2+y^2 < 16\}$, Range = $[\frac{1}{4}, \infty)$



Ex ③: $f(x, y) = \frac{y}{x^2}$

Domain = $\{(x, y) \in \mathbb{R}^2, x \neq 0\}$

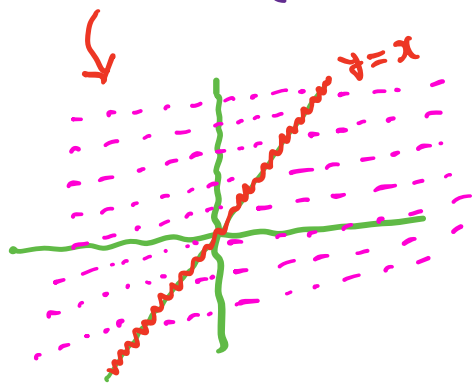


Range = All reals ($\mathbb{R}$)

$\mathbb{R} = (-\infty, \infty)$

Ex ④: $f(x, y) = \frac{1}{x-y}$

Domain = $\{(x, y) \in \mathbb{R}^2 : y \neq x\}$



Range = $(-\infty, 0) \cup (0, \infty)$
 $= \mathbb{R} \setminus \{0\}$



Exercises:

(1) $z = f(x, y) = \sin^{-1}(y-x)$

(2) $z = f(x, y) = \sqrt{9-x^2-y^2}$

(3) $w = f(x, y, z) = xy \ln z$

(4) $z = f(x, y) = \sqrt{y-x^2}$